Ejercicio 18: Resuelve

$a)\, 2^{3/10}.2^{3/5}.2$ $\qquad b)\,\frac{8^{-2/3}.8^2/3}{8^1/3}$ $\qquad c)\,\frac{\sqrt[3]{a^{5/7}}.\sqrt{a}}{\sqrt[5]{a^{2/3}}.\sqrt[4]{a^{2/5}}}$

$d)\,\frac{a^{2/5}.a^{7/5}}{a^{3/5}}$ $\qquad e)\,\frac{(10^2)^{-3}.(10^3)^{1/6}}{10.(10^4)^{-1/2}}$ $\qquad f)\,\frac{1}{a}\sqrt{\frac{a.b^2}{4}}+3b\sqrt{\frac{1}{4a}}-\frac{1}{a}\sqrt{ab^2}$

a) $2^{3/10}\cdot 2^{3/5}\cdot 2^{1} = 2^{\frac{3}{10}+\frac{3}{5}+1} = 2^{\frac{19}{10}}$

CA

$\frac{3}{10}+\frac{3}{5}+\frac{1}{1} = \frac{3+6+10}{10} = \frac{19}{10}$

b) $\frac{8^{-2/3}\cdot 8^{2/3}}{8^{1/3}} = 8^{-1/3} = (2^3)^{-1/3} = 2^{\frac{3}{1}\cdot\left(-\frac{1}{3}\right)} = 2^{-\frac{3}{3}}$

$\boxed{= 2^{-1} = \frac{1}{2}}$

$x^{n/m} = \sqrt[m]{x^n}$

$\sqrt[4]{(a^{2/5})^1} = (a^{2/5})^{1/4} = a^{\frac{2}{5}\cdot\frac{1}{4}}$

$= a^{2/20} = a^{1/10}$

c) $\frac{\sqrt[3]{(a^{5/7})}\cdot\sqrt{a}}{\sqrt[5]{(a^{2/3})^1}\cdot\sqrt[4]{(a^{2/5})^1}} = \frac{(a^{\frac{5}{7}})^{\frac{1}{3}}\cdot a^{1/2}}{(a^{\frac{2}{3}})^{1/5}\cdot(a^{\frac{2}{5}})^{1/4}} = \frac{a^{5/21}\cdot a^{1/2}}{a^{\frac{2}{3}\cdot\frac{1}{5}}\cdot a^{2/20}}$

$= \frac{a^{5/21}\cdot a^{1/2}}{a^{2/15}\cdot a^{1/10}} = \frac{a^{31/42}}{a^{7/30}} = a^{\frac{31}{42}-\frac{7}{30}} = a^{53/105}$

$\frac{5}{21}+\frac{1}{2} = \frac{10+21}{42} = \frac{31}{42}$

$\frac{31}{42}-\frac{7}{30} = \frac{930-294}{1260} = \frac{636}{1260} = \frac{53}{105}$

53
~~159~~ :3
~~318~~ :2
~~636~~ :2
~~1260~~ :2
~~630~~ :2
~~315~~ :3
105

$\frac{2}{15}+\frac{1}{10} = \frac{4+3}{30} = \frac{7}{30}$

Mínimo Común Multiplo (MCM)

MCM (15,10) = 3.5.2 = 30

15	3	10	2
5	5	5	5
1		1	

15 = 3.5 $\qquad$ 10 = 2.5

d) $\frac{a^{2/5}\cdot a^{7/5}}{a^{3/5}} = \frac{a^{9/5}}{a^{3/5}} = a^{\frac{9}{5}-\frac{3}{5}} = a^{6/5}$

$\frac{2}{5}+\frac{7}{5} = \frac{2+7}{5}$
$= \frac{9}{5}$

f) $f)\,\frac{1}{a}\sqrt{\frac{a.b^2}{4}}+3b\sqrt{\frac{1}{4a}}-\frac{1}{a}\sqrt{ab^2} =$

$= \frac{1}{a}\cdot\frac{\sqrt{a.b^2}}{\sqrt{4}} + 3b\,\frac{\sqrt{1}}{\sqrt{4a}} - \frac{1}{a}\cdot\frac{\sqrt{a}}{1}\cdot\frac{\sqrt{b^2}}{1} =$

$= \frac{1}{a}\cdot\frac{\sqrt{a}\cdot\sqrt{b^2}}{2} + 3b\cdot\frac{1}{2\sqrt{a}} - \frac{\sqrt{a}\cdot b}{a} =$

$= \frac{a^{1/2}\cdot b}{2a^1} + \frac{3b}{2\cdot a^{1/2}} - \frac{a^{1/2}\,b}{a} =$

$= a^{\frac{1}{2}-1}\cdot b + 3\cdot b\cdot a^{-1/2} - a^{1/2-1}\cdot b$

$\frac{1}{a}\cdot\frac{\sqrt{a}\cdot b}{2} = \frac{\sqrt{a}\cdot b}{2a}$

$\frac{3b}{2\sqrt{a}}\cdot\frac{\sqrt{a}}{\sqrt{a}} = \frac{3b\sqrt{a}}{2(\sqrt{a})^2} = \frac{3b\sqrt{a}}{2.a}$

$= \frac{3b.a^{1/2}}{} = \frac{3b\,a^{(1/2-1)}}{2}$

$$= \frac{a^{\frac{1}{2}-1}}{2}.b + \frac{3.b.a^{-1/2}}{2} - a^{1/2-1}.b$$

$2\sqrt{a} \quad \sqrt{a}$

$$= \frac{3b.a^{1/2}}{2a^1} = \frac{3ba^{(1/2-1)}}{2}$$

$$= \frac{a^{-1/2}b}{2} + \frac{3a^{-1/2}b}{2} - a^{-1/2}.b.1$$

$$\frac{1}{2} - \frac{1}{1} = \frac{1-2}{2} = -\frac{1}{2}$$

$$\frac{1}{2} + \frac{3}{2} - \frac{1}{1} = \frac{1+3-2}{2} = \frac{2}{2} = 1$$

$$= a^{-1/2}b\left(\frac{1}{2} + \frac{3}{2} - 1\right) = a^{-1/2}b.1 = a^{-1/2}b$$

$$= a^{-1/2}b = \frac{1}{a^{1/2}}.b = \boxed{\frac{b}{\sqrt{a}}}.\frac{\sqrt{a}}{\sqrt{a}} = \frac{b\sqrt{a}}{(\sqrt{a})^2} = \boxed{\frac{b\sqrt{a}}{a}}$$

Ejercicio 19: Resuelve las siguientes operaciones combinadas con números enteros.

$a)\ \left[\sqrt{(-6)^2-(-3)^2-5^2-2^0}\right] : [(-2)^3+8]$ $\qquad b)\ -\sqrt[3]{-24-3}-(1+5)^2-5^2\sqrt{10^4}:2$

$c)(-4+1)^3:(-3)^2+\sqrt{(-7)(-6)-(-12):(-2)}$ $\qquad d)\sqrt{\sqrt[3]{-8}+\sqrt{10^2-8^2}}+(-2)^2.(-2)^0.(-2)$

a) $\left[\sqrt{(-6)^2-(-3)^2-5^2-2^0}\right] : [(-2)^3+8] =$

$$= \left[\sqrt{36-(9)-25-1}\right] : [\underbrace{-8+8}_{=0}] = \sqrt{36-35} : 0 = 1:0 = \frac{1}{0} \quad \nexists$$

b) $-\sqrt[3]{-24-3} - (1+5)^2 - 5^2.\sqrt{10^4} : 2 =$

$$= -\sqrt[3]{-27} - 6^2 - 5^2.10^2 : 2$$

$$= -(-3) - 36 - 25.100 : 2$$

$$= 3 - 36 - 2500:2$$

$$= 3 - 36 - 1250 = \boxed{-1283}$$

$$\cancel{\sqrt{2^2-3} = \sqrt{2^2} - \sqrt{3}}$$

No se puede distribuir

$$\sqrt{(10^4)} = (10^4)^{1/2} = 10^{4/2} = 10^2$$

$$\sqrt{10^4} = \sqrt{10^2.10^2} = \sqrt{10^2}.\sqrt{10^2}$$

$$= 10.10$$

$$= 10^2 = 100$$

c) $(-4+1)^3:(-3)^2+\sqrt{(-7)(-6)-(-12):(-2)}$

$$= (-3)^3 : 9 + \sqrt{42 - 6}$$

$$= \frac{-27}{9} + \sqrt{36}$$

$$= -3 + 6 = \boxed{3}$$

$d)\ \sqrt{\sqrt[3]{-8}+\sqrt{10^2-8^2}}+(-2)^2.(-2)^0.(-2)$

$\sqrt[3]{-8} = -2$

$\sqrt{10^2-8^2} = \sqrt{100-64} = \sqrt{36} = 6$

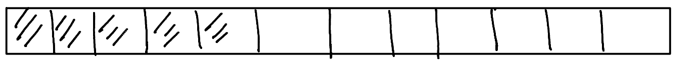
$= \sqrt{-2+6} + (-2)^{2+0+1}$

$= \sqrt{4} + (-2)^3 = 2 + (-8) = 2-8 = \boxed{-6}$

Ejercicio 20: Represente gráficamente las siguientes fracciones.

$a)\frac{5}{12}$ $\quad b)\frac{6}{10}$ $\quad c)\frac{6}{4}$ $\quad d)\frac{10}{3}$ $\quad b)\frac{6}{8}$

a)

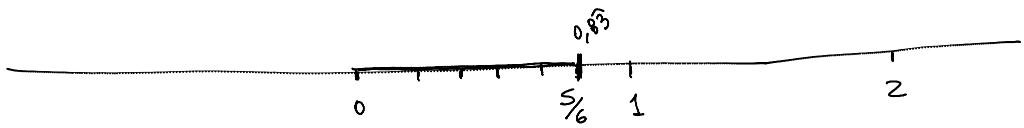


Ejercicio 21: Represente en la recta numérica los siguientes números racionales

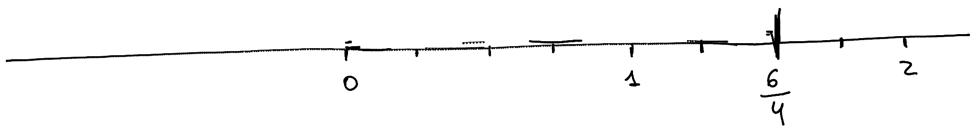
Números mixtos

$a)\frac{5}{6}$ $\quad b)\frac{6}{4}$ $\quad c)-\frac{3}{7}$ $\quad d)-2\frac{1}{3}$ $\quad e)-4\frac{(-4)}{5}$

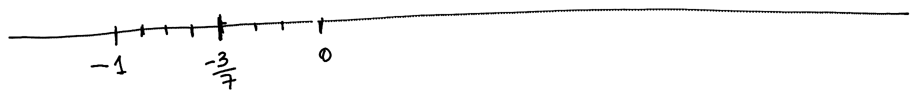
a)



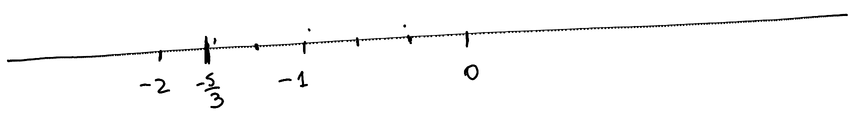
b)



c)



d) $2\frac{1}{3} = \frac{-2}{1} + \frac{1}{3} = \frac{-6+1}{3} = \frac{-5}{3}$



e) $-4\frac{-4}{5} = -4 + \frac{-4}{5} = -\frac{4}{1} - \frac{4}{5} = \frac{-20-4}{5} = \frac{-24}{5}$

Ejercicio 22: Compruebe si las siguientes fracciones son equivalentes

$a)\frac{2}{3}$ y $\frac{4}{6}$ $\quad b)\frac{1}{2}$ y $\frac{2}{4}$ $\quad c) -\frac{3}{7}$ y $\frac{6}{7}$ $\quad d)\frac{4}{7}$ y $2.\frac{1}{2}$

a) $\frac{2}{3}$ y $\frac{4}{6}$

$\frac{2.2}{3.2} = \frac{4}{6}$

Si son equivalentes $\frac{2}{3}$ y $\frac{4}{6}$

MÍNIMO COMÚN MÚLTIPLO

a) 15, 45, 90, 18

$$\begin{array}{r|l} 15 & 3 \\ 5 & 5 \\ 1 & \end{array} \qquad \begin{array}{r|l} 45 & 3 \\ 15 & 3 \\ 5 & 5 \\ 1 & \end{array} \qquad \begin{array}{r|l} 90 & 2 \\ 45 & 3 \\ 15 & 3 \\ 5 & 5 \\ 1 & \end{array} \qquad \begin{array}{r|l} 18 & 2 \\ 9 & 3 \\ 3 & 3 \\ 1 & \end{array}$$

$15 = 3.5$ $\quad 45 = 3^2.5$ $\quad 90 = 2.3^2.5$ $\quad 18 = 2.3^2$

$$\begin{aligned} mcm(15, 45, 90, 18) &= 2.3^2.5 \\ &= 2.9.5 \\ &= 90 \end{aligned}$$

90 es el número más pequeño que puede ser dividido por 15, 45, 90 y 18 a la vez.